



AE4238: Aero Engine Technology

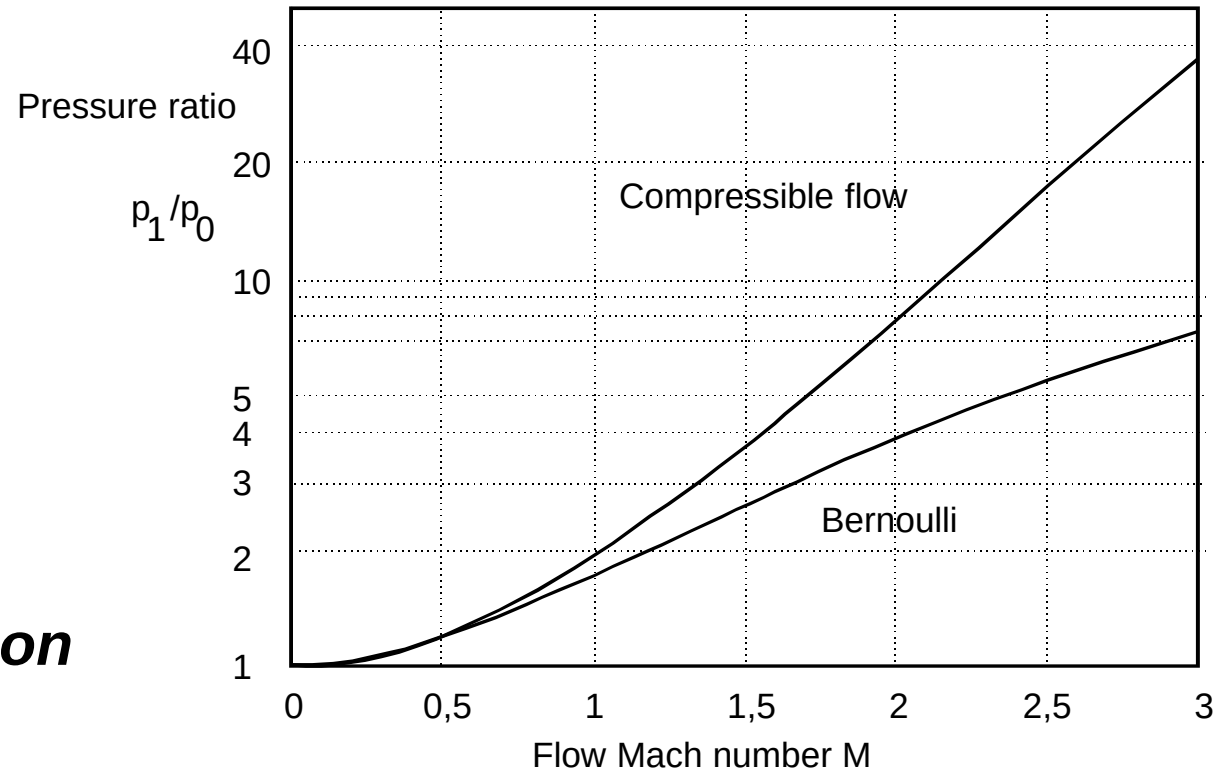
Engine Inlets-VI: Supersonic Inlets

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Compressibility

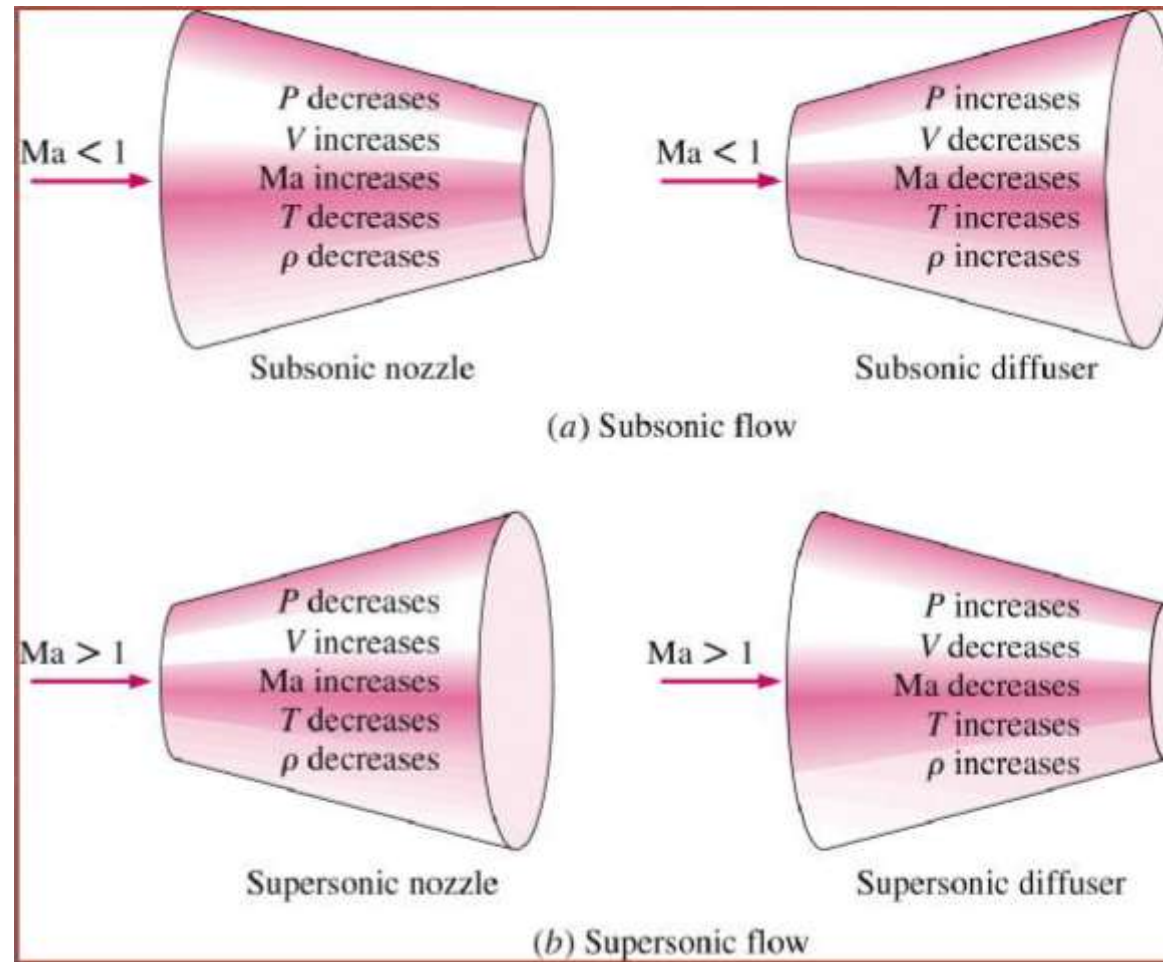
Difference in pressure rise in a diffuser between compressible and Bernoulli flow

Ram compression



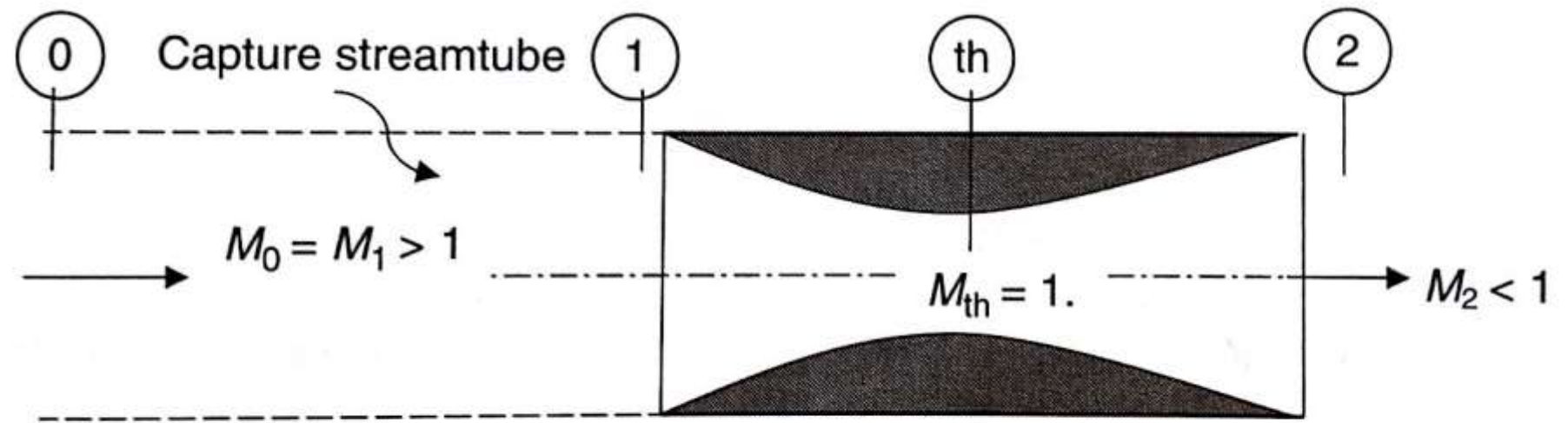
Compressibility

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$$dA/A = (M^2 - 1) * (dV/V)$$

An isentropic convergent-divergent inlet

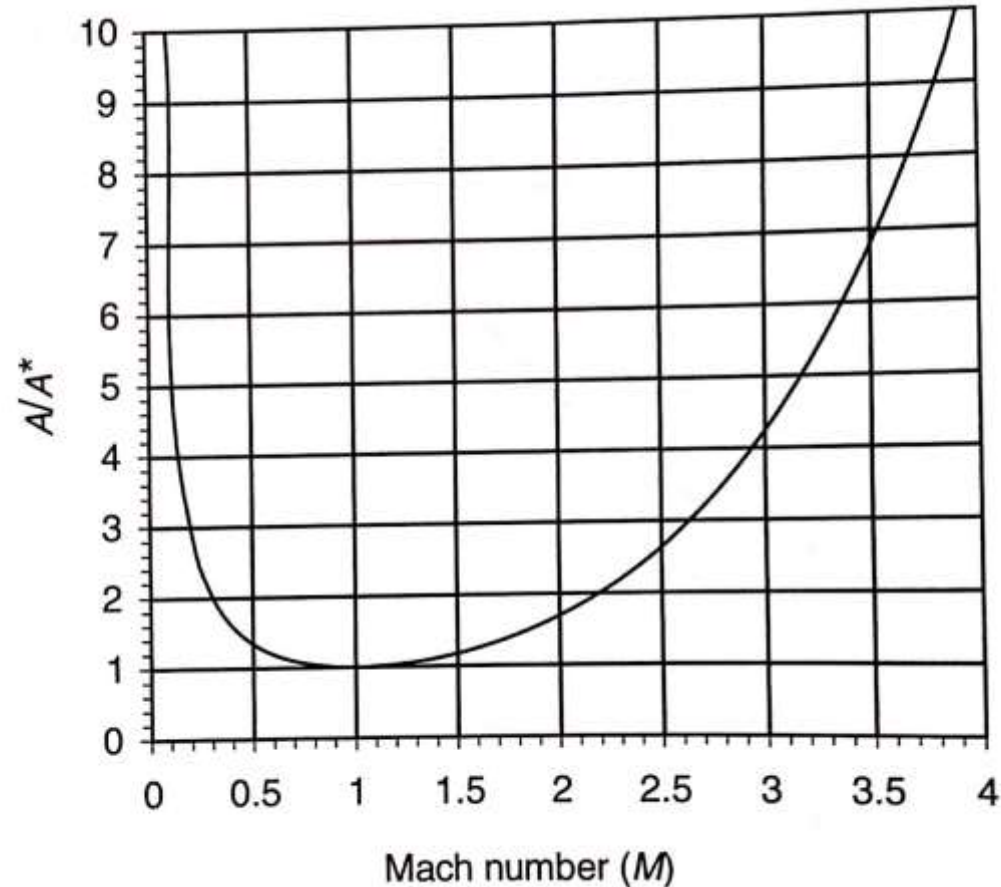


Source: Aircraft Propulsion, Farokhi

C-D area ratio

■ **FIGURE 5.27**
Isentropic area ratio as a
function of Mach
number ($\gamma = 1.4$) shows
that a C-D duct exhibits
choking behavior at a
supersonic as well as a
subsonic Mach number

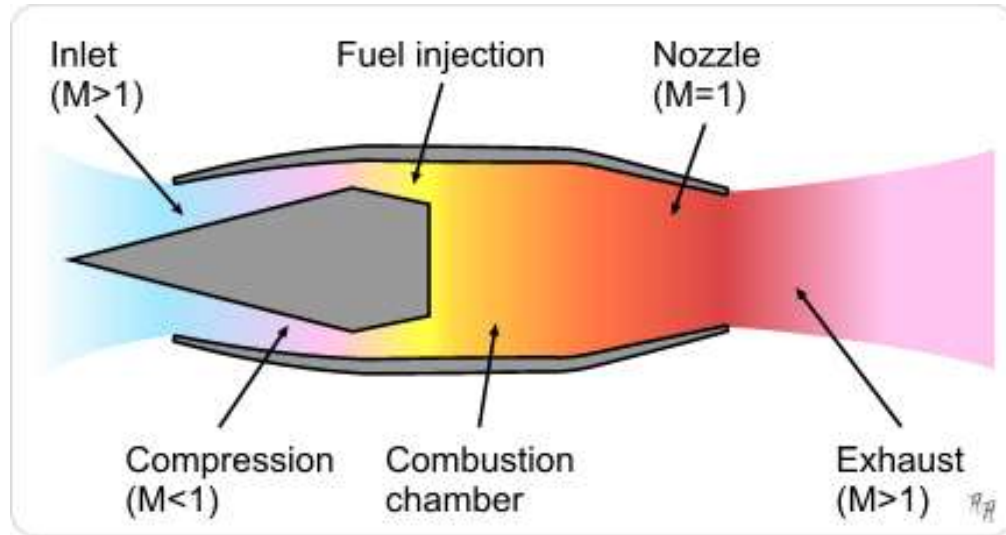
A^* is the area at
which the $M=1$



Source: Aircraft Propulsion, Farokhi

Ramjet Intake

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Source: Wikipedia

Inlet of SR-71

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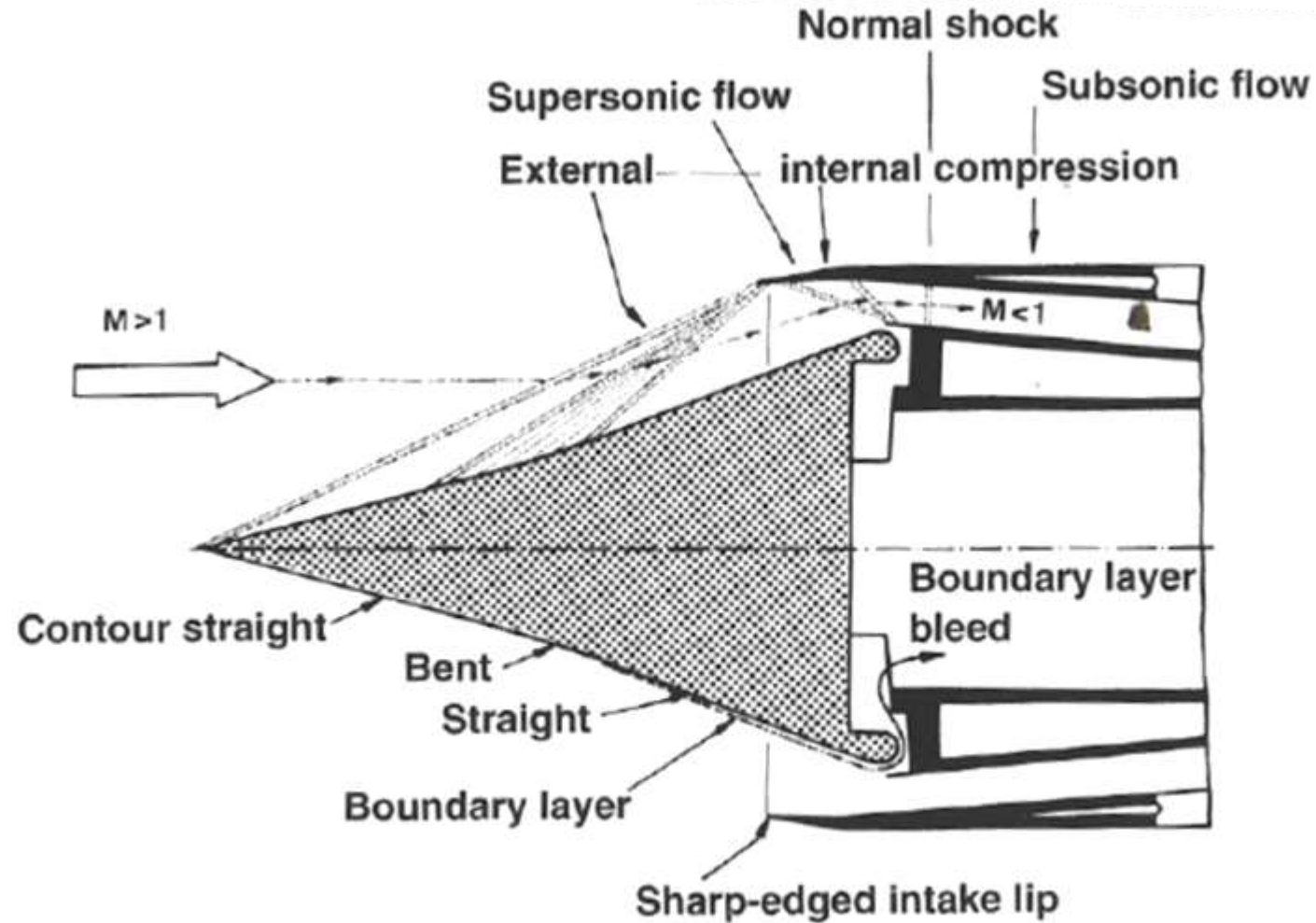
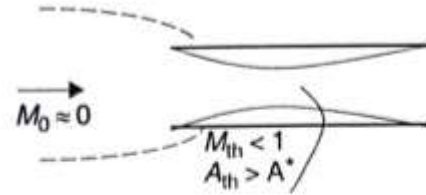


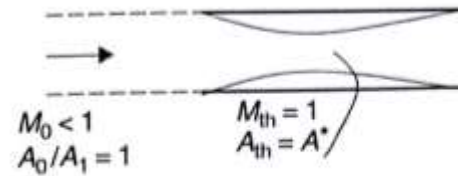
Fig 3-22 Mixed-compression inlet (Lockheed SR-71)

Flow characteristics of a CD inlet

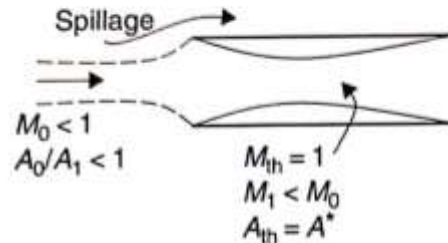
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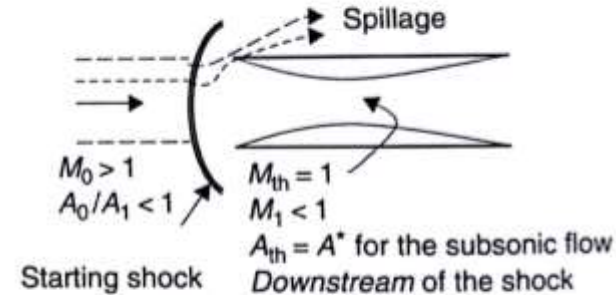
(a) At low speed the throat remains unchoked (note the capture ratio $A_0/A_1 > 1$)



(b) At a unique subsonic Mach number throat first chokes (note the capture ratio $A_0/A_1 = 1$)



(c) At higher subsonic Mach numbers than (b) the capture ratio falls below 1 and spillage occurs (note the throat remains choked)



(d) At supersonic Mach numbers a starting (bow) shock is formed in front of the inlet causing spillage drag while the throat remains choked

A → D

Low speed → High Speed

Source: Aircraft Propulsion,
Farokhi

Operation

Subcritical:

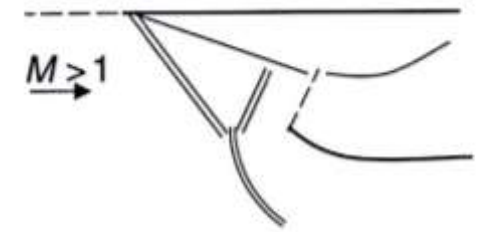
- Mach number below the design value.
- The normal shock occurs ahead of the cowl lip.
- High external drag due to spillage.

Critical:

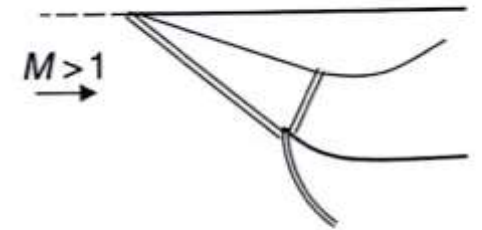
- Design point operation.
- The normal shock is located exactly at the cowl lip.

Supercritical:

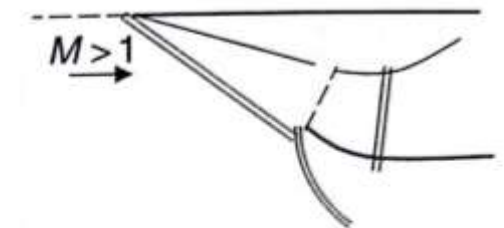
- Normal shock occurs in a region of higher Mach number
- Mass flow is the same as the critical mode.



(a) *Subcritical* mode with normal shock on the ramp



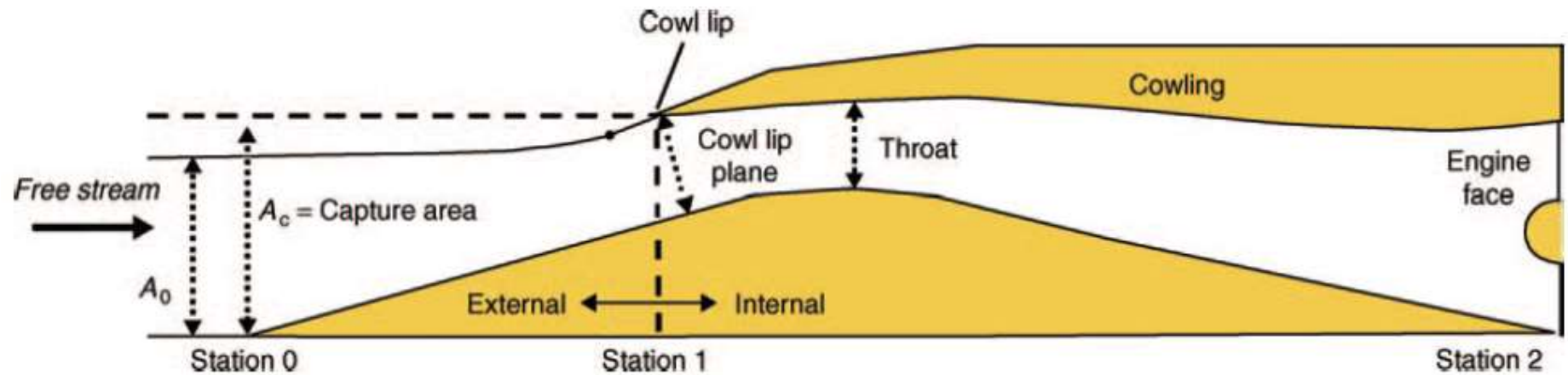
(b) *Critical* mode with N.S. at the throat



(c) *Supercritical* mode N.S. inside the duct

Supersonic Inlet

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Mixed compression Inlet

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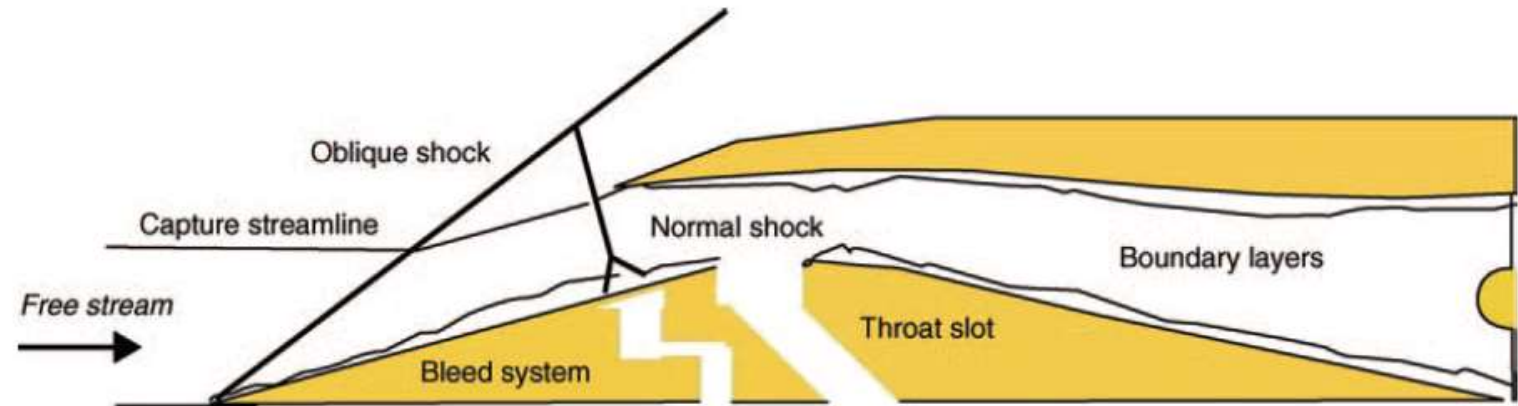


Figure 3. Supersonic external compression inlet with flow control.

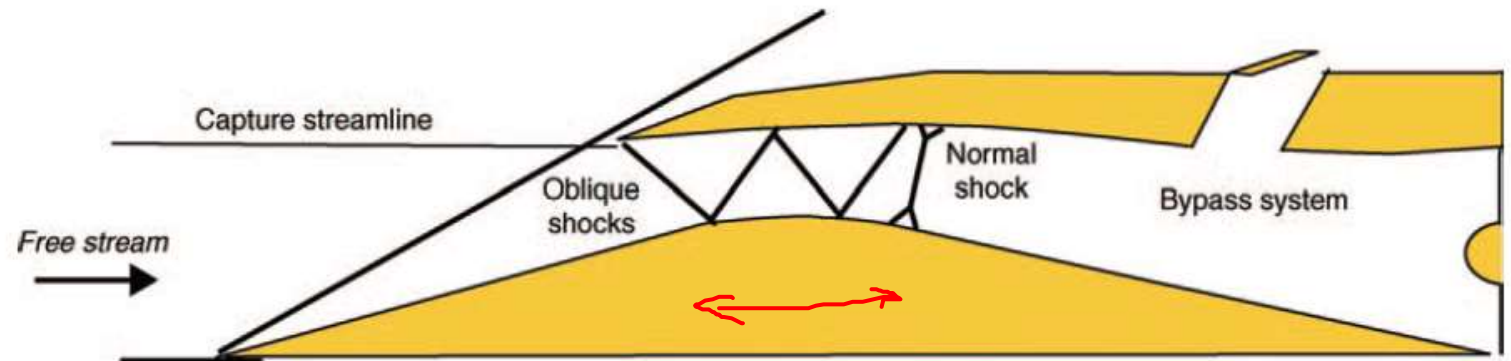
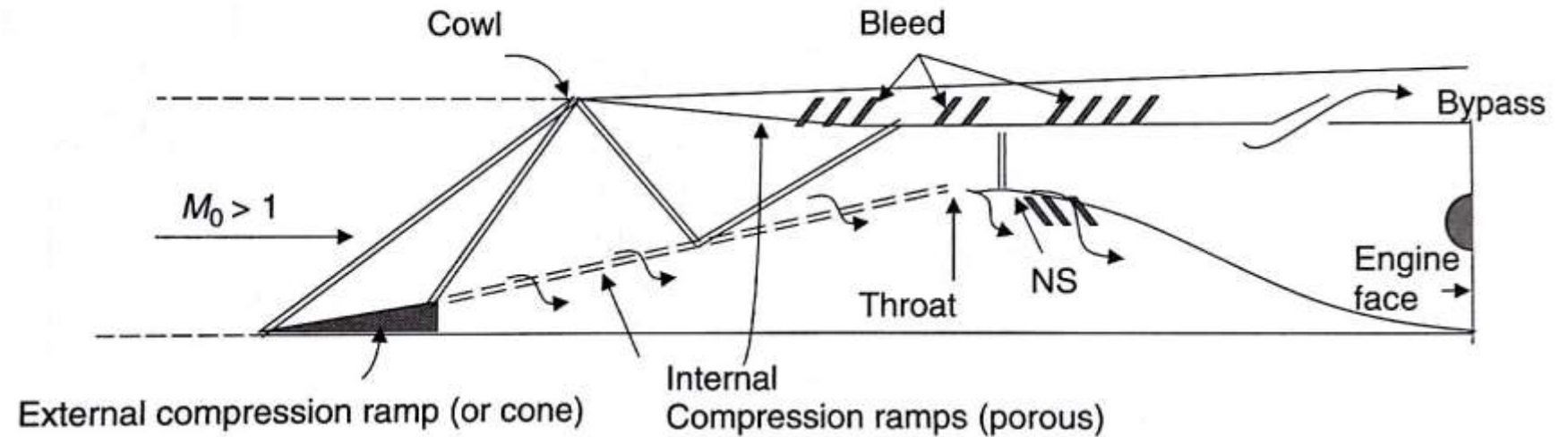


Figure 4. Supersonic mixed compression inlet with bypass system.

Flow characteristics of a CD inlet

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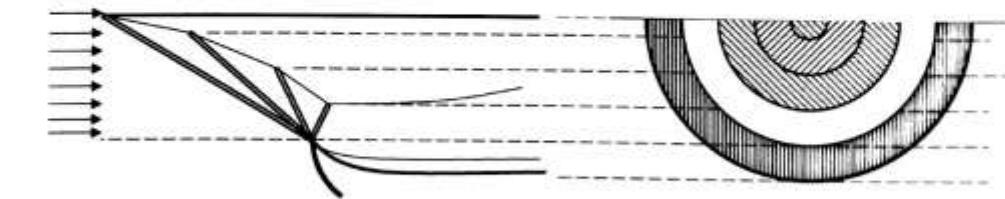
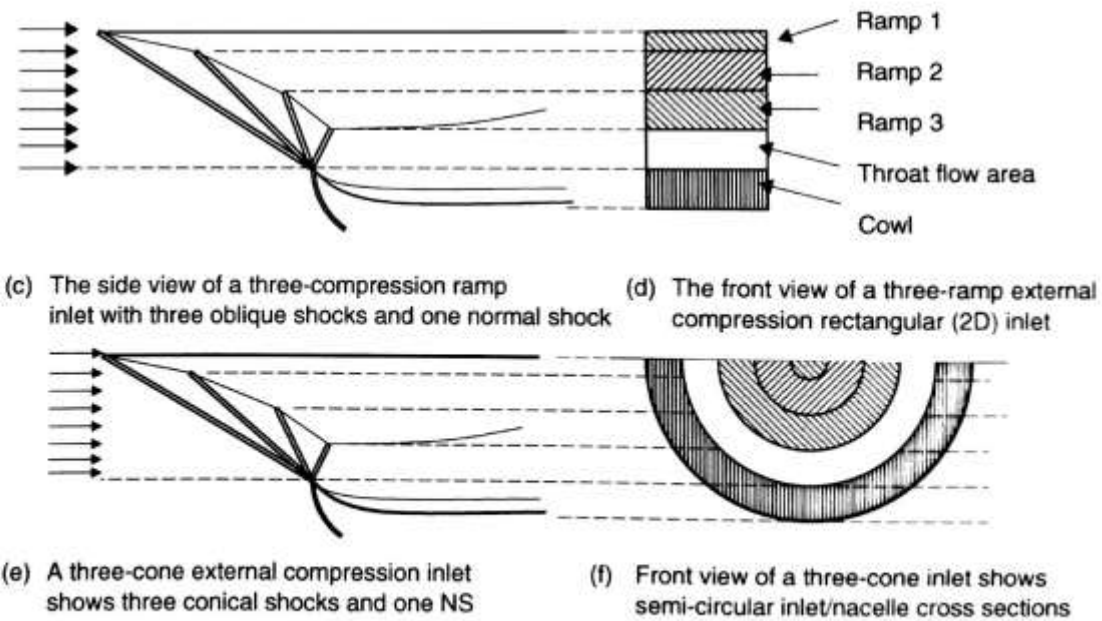
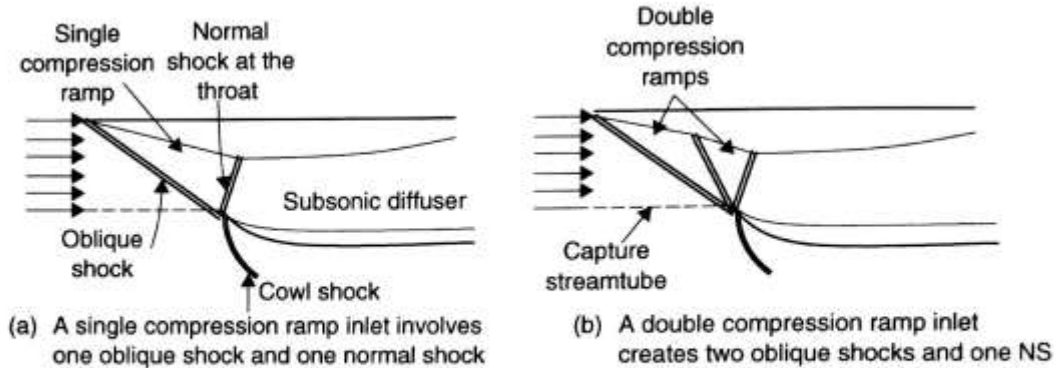
■ **FIGURE 5.46** A mixed compression inlet with external and internal shocks, an internal throat

Source: Aircraft Propulsion, Farokhi

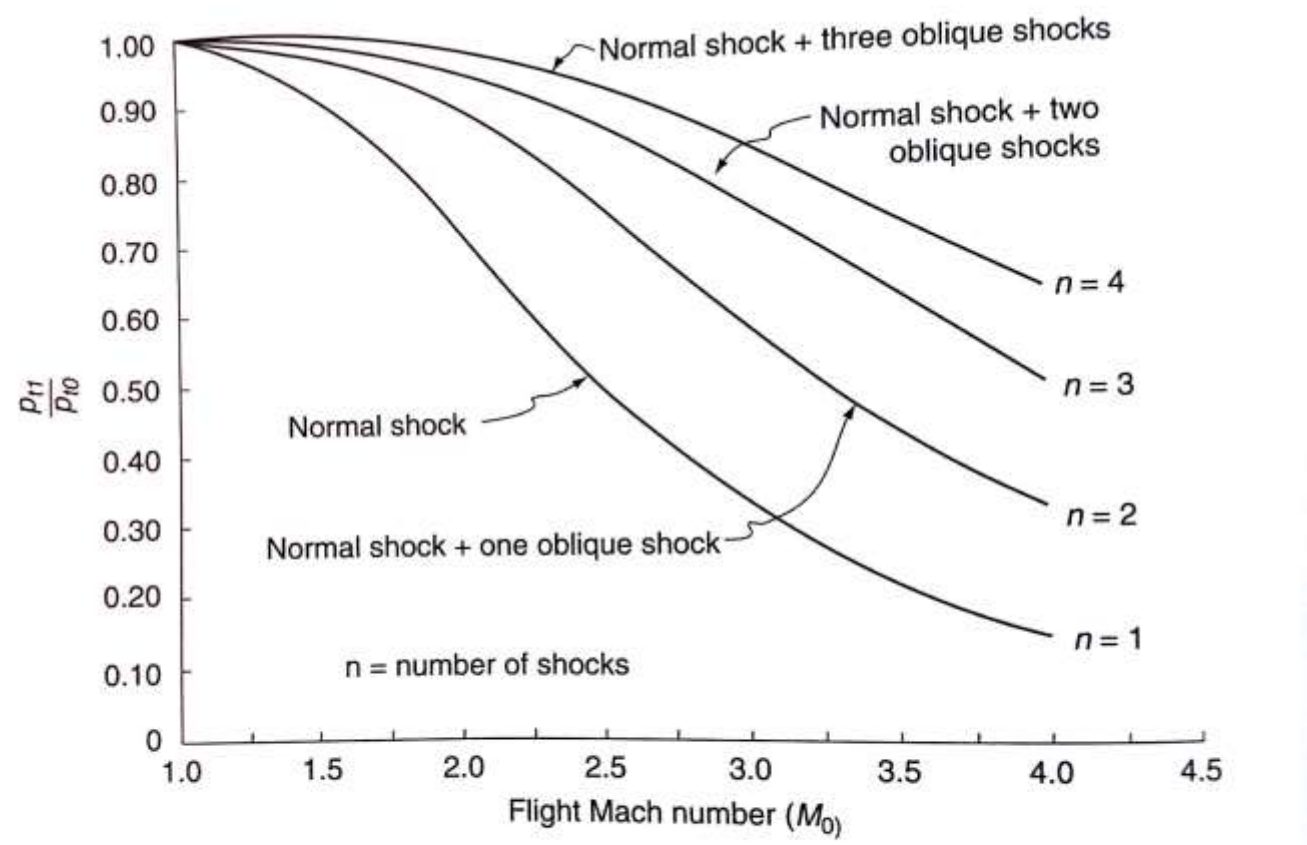
Annular Engine Intake (Mig 21)



■ **FIGURE 5.38**
Shock systems of two-dimensional and axisymmetric external compression inlets with the best backpressure placing the terminal (normal) shock at the lip (throat)



Flow characteristics of a CD inlet



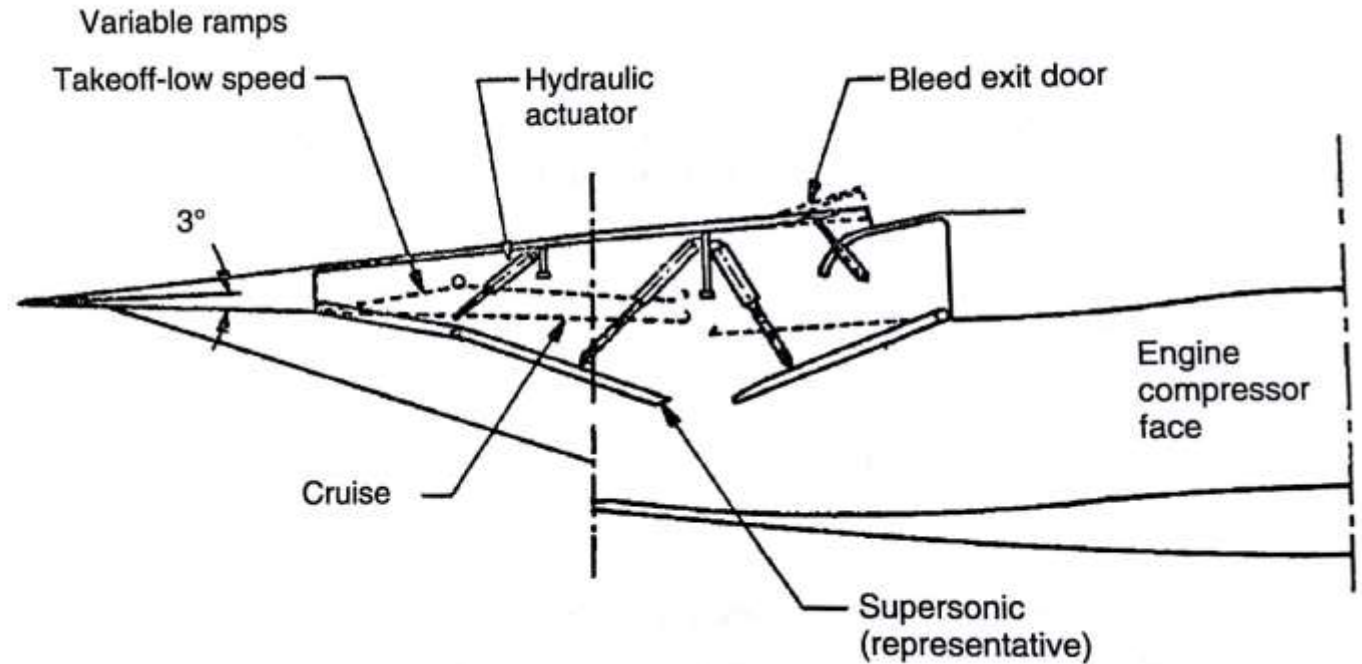
■ **FIGURE 5.40** Optimum total pressure recovery of external compression inlets (from Rodean, 1958.)

Source: Aircraft Propulsion, Farokhi

Flow characteristics of a CD inlet

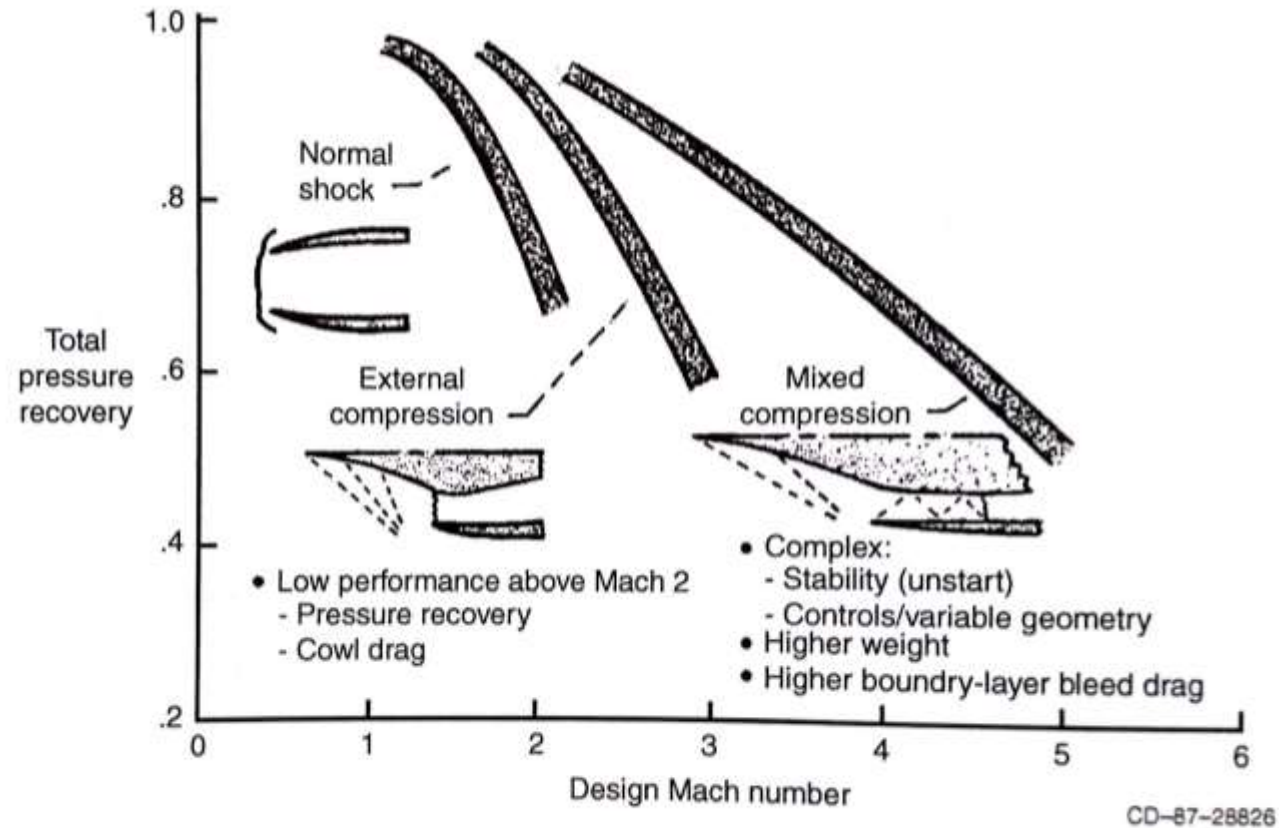
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■ **FIGURE 5.45**
Variable geometry inlet
shows ramp actuation at
different flight phases
and a bleed exit door
(from Abernethy and
Roberts, 1986)



Performance of Supersonic Inlet

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■ **FIGURE 5.48** Performance of supersonic inlets with flight Mach number (from 1987)

Source: Aircraft Propulsion, Farokhi

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Thank You



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